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```

```
data-engineering-workspace-prod

df_unwrap = (
    df_bronze_withColumn(
        "payload",
        try_parse_json(
            dynamic_json_to_standard_json_udf(to_json(col("payload"))).cast("string")
        ),
    )
)
# display(df_unwrap.limit(10))

df_bronze_with_paths = df_unwrap.withColumn(
    "response_body_path",
    when(
        expr("variant_get(payload, '$.NewImage.response_body', 'String')").rlike(
            r"^[a-z-0-9, \- \. ]+$"
        ),
        concat(
            lit(source_bucket_name),
            expr("variant_get(payload, '$.NewImage.response_body', 'String')"),
        ).otherwise(lit(None)),
    ).withColumn(
        "request_body_path",
        when(
            expr("variant_get(payload, '$.NewImage.request_body', 'String')").rlike(
                r"^[a-z-0-9, \- \. ]+$"
            ),
            concat(
                lit(source_bucket_name),
                expr("variant_get(payload, '$.NewImage.request_body', 'String')"),
            ).otherwise(lit(None)),
        ).otherwise(lit(None)),
    )
)
# display(df_bronze_with_paths.limit(10))
```

```
data-engineering-workspace-prod

request_body_paths_df = df_bronze_with_paths.select("request_body_path").distinct().filter(col("request_body_path").isNotNull())
print("Request Files", request_body_paths_df.count())
# Repartition for parallelism before collect
request_body_paths_df = request_body_paths_df.repartition(8)

# Use Spark to check file existence in bulk, avoiding Python loops
request_paths = [row.request_body_path for row in request_body_paths_df.collect()]
df_request_body_file = (
    spark.read.format("binaryfile")
    .option("pathGlobFilter", "**")
    .load(request_paths)
    .repartition(8)
    .select(
        col("path").alias("request_file_path"),
        col("content").cast("string").alias("request_body"),
        from_utc_timestamp(
            col("modification_timestamp"),
            "Asia/Kolkata"
        ).cast("timestamp_ntz").alias("request_file_modification_timestamp")
    )
)

df_request_body_file = df_request_body_file.withColumn(
    "request_body",
    from_json(
        col("request_body"),
        MapType(StringType(), StringType())
    )
)

# Identify not found paths by anti-join
found_paths_df = df_request_body_file.select("request_file_path").distinct()
not_found_paths_df = request_body_paths_df.join(
    request_body_paths_df.request_body_path == found_paths_df.request_file_path,
    "anti"
).select(col("request_body_path").alias("request_file_path"))

if not_found_paths_df.count() > 0:
    not_found_paths_df.withColumn("request_body", create_map(lit("error"), lit("file not found"))) \
        .withColumn("request_file_modification_timestamp", lit(None).cast("timestamp_ntz"))
```

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data-engineering-workspace-prod

br_pib_cal_hunter_v4

```
if not_found_paths_df.count() > 0:
    not_found_df = not_found_paths_df.withColumn("request_body", create_map(lit("error"), lit("file not found"))) \
        .withColumn("request_file_modification_timestamp_isit", lit(None)).cast("timestamp_ntz")
    df_request_body_file = df_request_body_file.unionByName(not_found_df).repartition(8)

if df_request_body_file.isEmpty():
    df_request_body_file = spark.createDataFrame([], schema="request_file_path STRING, request_body MAP<STRING,STRING>, request_file_modification_timestamp_isit timestamp_ntz")
# display(df_request_body_file)

response_body_paths_df = df_bronze_with_paths.select("response_body_path").distinct().filter(col("response_body_path").isNotNull())
print("Response Files:", response_body_paths_df.count())
# repartition for parallelism before collect
response_body_paths_df = response_body_paths_df.repartition(8)

# Use Spark to check file existence in bulk, avoiding Python loops
response_paths = [row.response_body_path for row in response_body_paths_df.collect()]
df_response_body_file = (
    spark.read.format("binaryfile")
    .option("pathFilter", "-*")
    .load(response_paths)
    .repartition(8)
    .select(
        col("path").alias("response_file_path"),
        col("content").cast("string").alias("response_body"),
        from_utc_timestamp(
            col("modification_time"),
            "AsiaKolkata"
        ).cast("timestamp_ntz").alias("response_file_modification_timestamp_isit")
    )
)

df_response_body_file = df_response_body_file.withColumn(
    "response_body",
    from_json(
        col("response_body"),
        MapType(StringType(), StringType())
    )
)
```

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data-engineering-workspace-prod

br_pib_cal_hunter_v4

```
df_response_body_file = df_response_body_file.withColumn(
    "response_body",
    from_json(
        col("response_body"),
        MapType(StringType(), StringType())
    )
)

# Identify not found paths by anti-join
found_response_paths_df = df_response_body_file.select("response_file_path").distinct()
not_found_response_paths_df = response_body_paths_df.join(
    found_response_paths_df,
    response_body_paths_df.response_body_path == found_response_paths_df.response_file_path,
    "left anti"
).select(col("response_body_path").alias("response_file_path"))

if not_found_response_paths_df.count() > 0:
    not_found_response_paths_df = not_found_response_paths_df.withColumn("response_body", create_map(lit("error"), lit("file not found"))) \
        .withColumn("response_file_modification_timestamp_isit", lit(None)).cast("timestamp_ntz")
    df_response_body_file = df_response_body_file.unionByName(not_found_response_paths_df).repartition(8)

if df_response_body_file.isEmpty():
    df_response_body_file = spark.createDataFrame([], schema="response_file_path STRING, response_body MAP<STRING,STRING>, response_file_modification_timestamp_isit timestamp_ntz")
# display(df_response_body_file)

df_joined = pipeline = (
    df_bronze_with_paths
    .join(
        df_request_body_file.hint("broadcast"),
        trim(lower(df_bronze_with_paths.request_body_path)) == trim(lower(df_request_body_file.request_file_path)),
        "left"
    )
    .join(
        df_response_body_file.hint("broadcast"),
        trim(lower(df_bronze_with_paths.response_body_path)) == trim(lower(df_response_body_file.response_file_path)),
        "left"
    )
)
```

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```
11
df_joined = pipeline = (
    df_bronze_with_paths
    .join(
        df_request_body_file.hint("broadcast"),
        trim(lower(df_bronze_with_paths.request_body_path)) == trim(lower(df_request_body_file.request_file_path)),
        "left"
    )
    .join(
        df_response_body_file.hint("broadcast"),
        trim(lower(df_bronze_with_paths.response_body_path)) == trim(lower(df_response_body_file.response_file_path)),
        "left"
    )
    .drop("request_file_path", "response_file_path")
    .withColumn("request_body", parse_json(to_json(col("request_body"))))
    .withColumn("response_body", parse_json(to_json(col("response_body"))))
)
# display(df_joined)
# display(df_joined.dtypes)

12
df_final = df_joined.toDF(*(c.lower().replace(" ", "_") for c in df_joined.columns))
duplicates = [item for item, count in Counter(df_final.columns).items() if count > 1]
if duplicates:
    raise Exception("Duplicate column found: (duplicates)")

13
df_final.write.format("delta") \
    .mode("overwrite") \
    .option("mergeSchema", "true") \
    .option("delta.columnMapping.mode", "name") \
    .option("optimize.write", "true") \
    .option("optimize.write.merge", "true") \
    .saveAsTable(bronze_table_full_2)
```

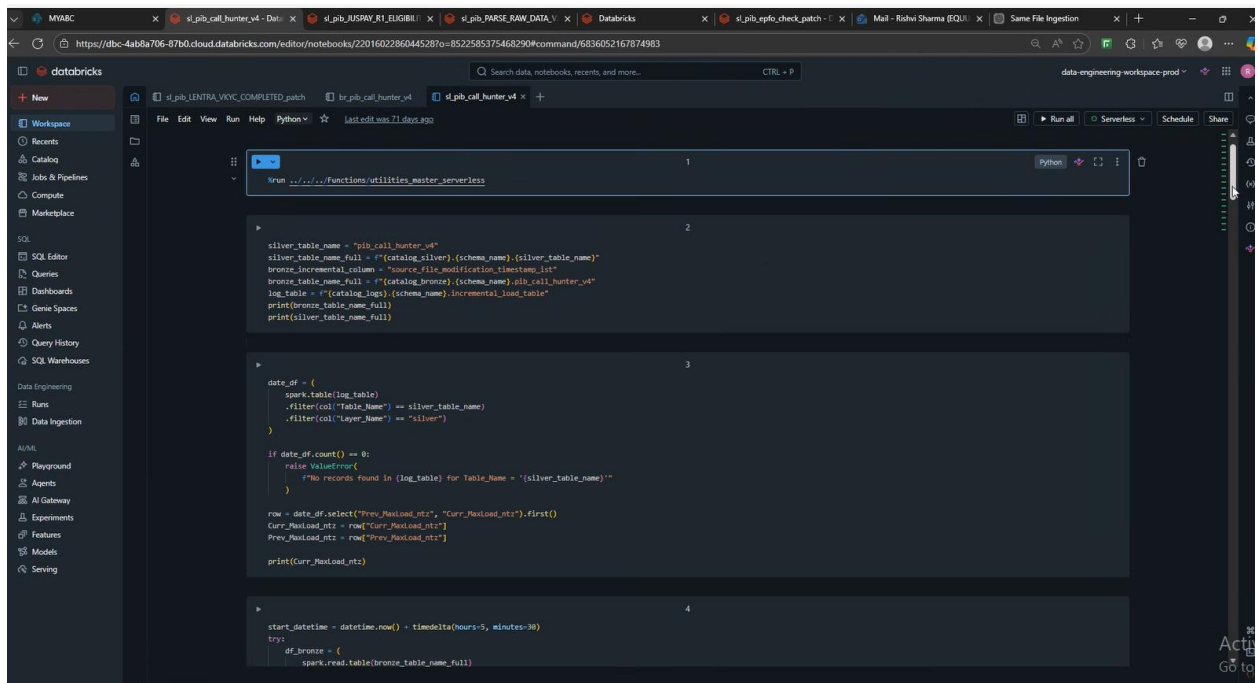
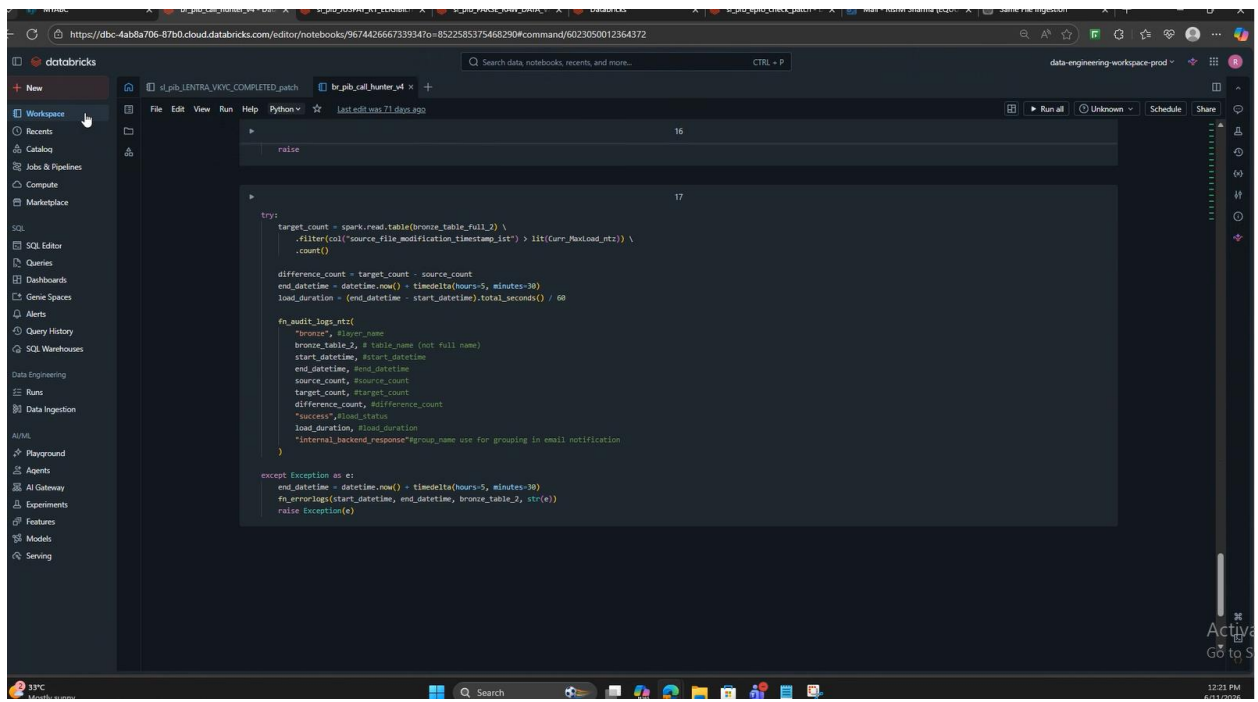
```
13

14
# Always attempt clustering (safe to run multiple times)
spark.sql(f"ALTER TABLE {bronze_table_full_2} CLUSTER BY AUTO")
# display(spark.sql(f"describe {bronze_table_full_2}"))

15
try:
    fn_update_incremental_load_table_previous_ntz(
        table_name=bronze_table_2, layer_name="bronze", schema="digital_ios"
    )
except Exception as e:
    end_datetime = datetime.now() + timedelta(hours=5, minutes=30)
    fn_errorlogs(start_datetime, end_datetime, bronze_table_2, str(e))
    raise

16
try:
    fn_update_incremental_load_table_ntz(
        catalog=catalog_bronze,
        schema="digital_ios",
        table=bronze_table_2,
        table_name=bronze_table_2,
        layer_name="bronze",
        col_name="source_file_modification_timestamp_int",
    )
except Exception as e:
    end_datetime = datetime.now() + timedelta(hours=5, minutes=30)
    fn_errorlogs(start_datetime, end_datetime, bronze_table_2, str(e))
    raise

17
try:
```



data-engineering workspace prod

```
keys_payload_values = keys["payload_keys"]

# Step 3: Extract values using variant_get
p_cols = [
    expr("variant_get(payload, 's.k', 'VARIANT')).alias('k')")
    for k in keys_payload_values
]

# Step 4: Select expanded columns
df_split = df_bronze.select("k", *p_cols)

# Step 5: Drop unwanted columns
df_split = df_split.drop(
    "payload",
)

# Step 1: Extract all dynamic keys
df_keys_2 = df_split.select(
    explode(map_keys(col("NewImage").cast("mapstring, variant"))).alias("NewImage_key"),
    explode(map_keys(col("Metadata").cast("mapstring, variant"))).alias("Metadata_key"),
    explode(map_keys(col("Keys").cast("mapstring, variant"))).alias("Keys_key"),
)

# Step 2: Collect distinct keys
keys_2 = df_keys_2.agg(
    collect_set("NewImage_key").alias("NewImage_keys"),
    collect_set("Metadata_key").alias("Metadata_keys"),
    collect_set("Keys_key").alias("Keys_keys"),
).collect()[0]

keys_NewImage_values = keys_2["NewImage_keys"]
keys_Metadata_values = keys_2["Metadata_keys"]
keys_Keys_values = keys_2["Keys_keys"]

# Step 3: Extract values using variant_get
key_cols = [
    expr("variant_get(Keys, 's.k', 'VARIANT')).alias('key_k')")
    for k in keys_Keys_values
]

md_cols = [
    expr("variant_get(Metadata, 's.k', 'VARIANT')).alias('md_k')")
    for k in keys_Metadata_values
]

nl_cols = [
    expr("variant_get(NewImage, 's.k', 'VARIANT')).alias('nl_k')")
    for k in keys_NewImage_values
]

# Step 4: Select expanded columns
df_split_2 = df_split.select("k", *key_cols, *md_cols, *nl_cols)

# Step 5: Drop unwanted columns
df_split_2 = df_split_2.drop(
    "Keys",
    "Metadata",
    "NewImage",
    "OldImage",
    "nl_response_body", # same as response_body_path
    "nl_request_body", # same as request_body_path
    "nl_endpoint", # same as endpoint
    "nl_application_id", # same as application_id
)

def ensure_nullable_column(df, src_col=None, target_col=None, dtype="string"):
    if src_col and src_col in df.columns:
        return df.withColumn(target_col, col(src_col).cast(dtype))
    else:
        return df.withColumn(target_col, lit(None).cast(dtype))
```

data-engineering workspace prod

```
key_cols = [
    expr("variant_get(Keys, 's.k', 'VARIANT')).alias('key_k')")
    for k in keys_Keys_values
]

md_cols = [
    expr("variant_get(Metadata, 's.k', 'VARIANT')).alias('md_k')")
    for k in keys_Metadata_values
]

nl_cols = [
    expr("variant_get(NewImage, 's.k', 'VARIANT')).alias('nl_k')")
    for k in keys_NewImage_values
]

# Step 4: Select expanded columns
df_split_2 = df_split.select("k", *key_cols, *md_cols, *nl_cols)

# Step 5: Drop unwanted columns
df_split_2 = df_split_2.drop(
    "Keys",
    "Metadata",
    "NewImage",
    "OldImage",
    "nl_response_body", # same as response_body_path
    "nl_request_body", # same as request_body_path
    "nl_endpoint", # same as endpoint
    "nl_application_id", # same as application_id
)

def ensure_nullable_column(df, src_col=None, target_col=None, dtype="string"):
    if src_col and src_col in df.columns:
        return df.withColumn(target_col, col(src_col).cast(dtype))
    else:
        return df.withColumn(target_col, lit(None).cast(dtype))
```

MYABC x sl_pib_cal_hunter_v4 - Data x sl_pib_JUSPAY_R1_ELIGIBILI x sl_pib_PARSE_RAW_DATA x Databricks x sl_pib_epfo_check_patch x Mail - Rishvi Sharma (EQU... x Same File Ingestion x + -

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Workspace

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8

```
df_split_2 = ensure_nullable_column(
    df=df_split_2,
    src_col="ml_status",
    target_col="ml_status",
    dtype="variant"
)
```

9

```
df_final = {
    df_split_2.withColumn(
        "inserted_timestamp_list",
        from_utc_timestamp(current_timestamp(), "Asia/Kolkata").cast("timestamp_ntz"),
    ),
    .withColumn(
        "bronze_inserted_timestamp_list",
        col("inserted_timestamp_list").cast("timestamp_ntz"),
    ),
    .selectExpr(
        "application_id",
        "partner_journey_id",
        "endpoint",
        "ml_status:string",
        "source_file_modification_timestamp_list", # file creation time
        "bronze_inserted_timestamp_list", # bronze time
        "inserted_timestamp_list", # silver time
        "EXCEPT(application_id, partner_journey_id, endpoint, ml_status, source_file_modification_timestamp_list, bronze_inserted_timestamp_list, inserted_timestamp_list)",
    )
}
df_final = df_final.withColumnRenamed(
    (col: col.replace("ml_", "")) for col in df_final.columns if col.startswith("ml_")
)

10
```

```
df_final = ensure_nullable_column(
    df=df_final,
    src_col="response_body",
    target_col="response_body",
    dtype="variant"
)
```

11

```
df_final = df_final.withColumn(
    "response_body",
    CASE
        WHEN response_body IS NULL THEN NULL
        WHEN try_parse_json(regexp_replace(response_body, '\\\\\\\\', '\\')) IS NULL
        THEN parse_json(
            to_json(
                named_struct(
                    'type', 'HTML',
                    'content', regexp_replace(response_body, '\\\\\\\\', '\\')
                )
            )
        )
        ELSE parse_json(regexp_replace(response_body, '\\\\\\\\', '\\'))
    END
)
df_final = df_final.withColumn(
    "request_body",
    CASE
        WHEN request_body IS NULL THEN NULL
        WHEN try_parse_json(regexp_replace(request_body, '\\\\\\\\', '\\')) IS NULL
        THEN parse_json(
            to_json(
                named_struct(
                    'type', 'HTML',
                    'content', regexp_replace(request_body, '\\\\\\\\', '\\')
                )
            )
        )
        ELSE parse_json(regexp_replace(request_body, '\\\\\\\\', '\\'))
    END
)
```

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MYABC x sl_pib_cal_hunter_v4 - Data x sl_pib_JUSPAY_R1_ELIGIBILI x sl_pib_PARSE_RAW_DATA x Databricks x sl_pib_epfo_check_patch x Mail - Rishvi Sharma (EQU... x Same File Ingestion x + -

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+ New

Workspace

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10

```
df_final = ensure_nullable_column(
    df=df_final,
    src_col="response_body",
    target_col="response_body",
    dtype="variant"
)
```

11

```
df_final = df_final.withColumn(
    "response_body",
    CASE
        WHEN response_body IS NULL THEN NULL
        WHEN try_parse_json(regexp_replace(response_body, '\\\\\\\\', '\\')) IS NULL
        THEN parse_json(
            to_json(
                named_struct(
                    'type', 'HTML',
                    'content', regexp_replace(response_body, '\\\\\\\\', '\\')
                )
            )
        )
        ELSE parse_json(regexp_replace(response_body, '\\\\\\\\', '\\'))
    END
)
df_final = df_final.withColumn(
    "request_body",
    CASE
        WHEN request_body IS NULL THEN NULL
        WHEN try_parse_json(regexp_replace(request_body, '\\\\\\\\', '\\')) IS NULL
        THEN parse_json(
            to_json(
                named_struct(
                    'type', 'HTML',
                    'content', regexp_replace(request_body, '\\\\\\\\', '\\')
                )
            )
        )
        ELSE parse_json(regexp_replace(request_body, '\\\\\\\\', '\\'))
    END
)
```

33°C Mostly sunny 12:28 PM 6/11/2026

```

-- Cell 11
CASE
  WHEN request_body IS NULL THEN NULL
  WHEN try_parse_json(regexp_replace(request_body, '\\\\\\\\', '\\')) IS NULL
  THEN parse_json(
    to_json(
      named_struct(
        'type', 'HTML',
        'content', regexp_replace(request_body, '\\\\\\\\', '\\')
      )
    )
  )
  ELSE parse_json(regexp_replace(request_body, '\\\\\\\\', '\\'))
END
---

-- Cell 12
# Step 1: Get all top-level keys from response_body
all_keys = (
  of_final.filter("response_body IS NOT NULL")
  .select(explode(map_keys(col(response_body))).cast("mapstring, variant").)$.alias("key"))
  .agg(collect_set("key").alias("keys"))
  .collect()[0][0]["keys"]
)

# Step 2: Dynamically detect fields that are STRING-type variants containing valid JSON objects/arrays
open_brace = chr(123) # '{' - avoids f-string escaping issues
check_exprs = [
  expr(""""
    MAX(CASE
      WHEN schema_of_variant(response_body.`k`) = 'STRING'
      AND LEPT(CAST(response_body.`k`) AS STRING), 1) IN ('{open_brace}', '{')
      AND try_parse_json(CAST(response_body.`k`) AS STRING) IS NOT NULL
      THEN 1 ELSE 0 END
    ) as `k`
  """)
  for k in all_keys
]

```

The screenshot displays a Databricks workspace interface. The top navigation bar shows the workspace name 's_pib_cal_hunter_v4' and the file 's_pib_PARSE_RAW_DATA'. The main editor area contains a Python script using PySpark. The script filters for non-null response bodies, parses them into proper JSON, and then uses a CASE statement to handle nested JSON structures. It also includes a check for duplicate columns and a final write operation.

```
12
result = df_final.filter(response_body IS NOT NULL).select("check_expch.collect()")
json_string_fields = [k for k in all_keys if result[k] == 1]
print("Detected stringified JSON fields: (json_string_fields)")

# Step 3: Parse each stringified JSON field into proper JSON within response_body
for field in json_string_fields:
    df_final = df_final.withColumn(
        "response_body",
        expr("""
            CASE
            WHEN response_body IS NULL THEN Null
            WHEN response_body.'field' IS NOT NULL
            AND try_parse_json(CAST(response_body.'field' AS STRING)) IS NOT NULL THEN
                parse_json_to_json_concat(
                    map_filter(
                        CAST(response_body AS MAP<STRING, VARIANT>),
                        (k, v) -> k != 'field'
                    ),
                    map(
                        'field',
                        try_parse_json(CAST(response_body.'field' AS STRING))
                    )
                )
            ELSE response_body
            END
        """))

13
# display(df_final.dtypes)
df_final = df_final.toDF(*(c.lower().replace(" ", "_") for c in df_final.columns))
duplicates = [item for item, count in Counter(df_final.columns).items() if count > 1]
if duplicates:
    raise Exception("Duplicate columns found: {duplicates}")

14
df_final.write.format("delta") \
```



```
13
# display(df_final.dtypes)
df_final = df_final.toDF([c.lower().replace(" ", "_") for c in df_final.columns])
duplicates = [item for item, count in Counter(df_final.columns).items() if count > 1]
if duplicates:
    raise Exception(f"Duplicate columns found: {duplicates}")

14
df_final.write.format("delta") \
    .mode("append") \
    .option("mergeSchema", "true") \
    .option("delta.columnMapping.mode", "name") \
    .saveAsTable(silver_table_name_full)

# Always attempt clustering (safe to run multiple times)
spark.sql(f"ALTER TABLE {silver_table_name_full} CLUSTER BY AUTO")

15
try:
    fn_update_incremental_load_table_previous_ntz(
        table_name=silver_table_name,
        layer_name="silver",
        schema="digital_ios"
    )
except Exception as e:
    end_datetime = datetime.now() + timedelta(hours=5, minutes=30)
    fn_error_logs(start_datetime, end_datetime, silver_table_name, str(e))
    raise

16
try:
    fn_update_incremental_load_table_ntz(
        catalog=catalog_silver,
        schema="digital_ios",
        table_name=silver_table_name,
        layer_name="silver",
        col_name="source_file_modification_timestamp"
    )
except Exception as e:
    end_datetime = datetime.now() + timedelta(hours=5, minutes=30)
    fn_error_logs(start_datetime, end_datetime, silver_table_name, str(e))
    raise

17
try:
    target_count = (
        spark.read.table(silver_table_name_full)
        .filter(col("source_file_modification_timestamp") > lit(Curr_MaxLoad_ntz))
        .count()
    )
    difference_count = target_count - source_count
    end_datetime = datetime.now() + timedelta(hours=5, minutes=30)
    load_duration = (end_datetime - start_datetime).total_seconds() / 60

    fn_email_logs_ntz(
        "silver", # layer_name
        silver_table_name, # table_name (not full name)
        start_datetime, # start_datetime
        end_datetime, # end_datetime
        source_count, # source_count
        target_count, # target_count
        difference_count, # difference_count
        "success", # load_status
        load_duration, # load_duration
        "internal_backend_response", # group_name use for grouping in email notification
    )
except Exception as e:
```

```
16
try:
    fn_update_incremental_load_table_ntz(
        catalog=catalog_silver,
        schema="digital_ios",
        table_name=silver_table_name,
        layer_name="silver",
        col_name="source_file_modification_timestamp"
    )
except Exception as e:
    end_datetime = datetime.now() + timedelta(hours=5, minutes=30)
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17
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        target_count, # target_count
        difference_count, # difference_count
        "success", # load_status
        load_duration, # load_duration
        "internal_backend_response", # group_name use for grouping in email notification
    )
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```

MYABC x sl_pib_cal_hunter_v4 - Dat x sl_pib_AUSRAY_R1_EUGIBH x sl_pib_PARSE_RAW_DATA x Databricks x sl_pib_erro_check_patch x Mail - Rohini Sharma (SQU x Same File Ingestion x

https://dbc-4ab8a706-87b0.cloud.databricks.com/edit/notebooks/220160228604452870-8522585375468290?command=6836052167874983

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+ New Workspace File Edit View Run Help Python Last edit was 71 days ago Run all Serverless Schedule Share

16

```
raise
```

17

```
try:
    target_count = (
        spark.read.table(silver_table_name_full)
        .filter(col("source_file_modification_timestamp") > lit(Curr_Maxload_mtz))
        .count()

    difference_count = target_count - source_count
    end_datetime = datetime.now() - timedelta(hours=5, minutes=30)
    load_duration = (end_datetime - start_datetime).total_seconds() / 60

    # send_email_logs_mtz(
    #     "silver", # layer name
    #     silver_table_name, # table name (not full name)
    #     start_datetime, # start_datetime
    #     end_datetime, # end_datetime
    #     source_count, # source_count
    #     target_count, # target_count
    #     difference_count, # difference_count
    #     "success", # load_status
    #     load_duration, # load_duration
    #     "internal_backend_response", # group_name use for grouping in email notification
    # )

except Exception as e:
    end_datetime = datetime.now() - timedelta(hours=5, minutes=30)
    # send_email_logs_mtz(start_datetime, end_datetime, silver_table_name, str(e))
    raise Exception(e)
```

Activate
Go to Settings